frequency resolution, 173-176
inverse DFT, 152-156, 573-580
Discrete Fourier transform (continued)
negative frequencies, 196-200, 209-211, 226-227, 569-575
orthognal basis functions, 158-159
periodic frequency domain, 196-200
periodic time domain, 194-196
real DFT, 141-160, 225-227, 519-521
spectral density, 156
spectral leakage, 175-176
synthesis, 152-156, 573-580
Discrete Time Fourier Transform (DTFT). *See* Fourier transform
Distant neighbors in images, 439
Dithering, 38-39, 374
Division of frequency domain signals, 181-182
DN (digital number) in images, 374
Dolby stereo, 362
Domain (defined), 12
Double precision, 70-74, 284, 339
DSP microprocessor, 84
DTFT. *See* Fourier transform
Duality, 161, 210-212, 236
Dynamic range, 261-262, 378

Ear, 351-355
Echo control in telephones, 5
Echo location, 7-8. *See also* Radar; Sonar; Seismology
Echoes in music, 5
Edge detection, 402-403, 416-417
Edge response, 426-430
EFM (8 to 14 modulation), 360
Electric circuit analysis
phasor transform method, 563-566
Laplace transform method, 592-599
Electroencephalogram (EEG), 292-293
End effect. *See under* Convolution
Erosion, 435. *See also* Morphological processing
Euler's relation. *See under* Complex numbers
Even field in video, 384-385
Even/odd decomposition, 102, 240-242
Even symmetry, 102, 196, 240-241
Evolution, neural network learning, 470
Execution speed. *See* Speed
Exponential, two ways to generate, 606-607
Eye, 376-381, 404, 409, 432-434

False-positive (false-negative), 453-454
Fast Fourier transform (FFT), 180, 225-242
Feature extraction, 458
FFT convolution, 140, 179-184, 311-318, 411, 416-422
Field, television, 384-386
Filters, analog
See also Filters, digital; Filters, recursive
in ADC and DAC, 48-59
antialias, 48-49, 55-59, 172-173
Bessel, 49-59, 330, 361
Butterworth, 49-58, 600-604
Chebyshev, 49-58, 600-604
design methods, 49-59, 600-604
digital, compared to, 261-262, 343-345
elliptic, 54
frequency response, 52-54, 176-179, 598-599
high-pass, 51
low-pass, 49-54, 322, 343-345, 600-604
notch filter, 519-520, 592-597
overshoot, step response, 54-55
pulse response, 55
reconstruction filter for DAC, 44-49, 480
ringing, step response, 54-55
roll-off, 52-53
Sallen-key circuit, 49-50, 600-601
smoothing, 322
stability of analog filters, 541, 600-601
step response, 54-55, 322-323
switch-capacitor, 51-52
Filters, digital
See also Filters, analog; Filters, recursive
band-pass, 177-179, 268-274, 293
band-reject, 268-274, 291, 293
custom response, 297-310
cutoff frequency (defined), 268
edge enhancement, 400-401
even order filter, 603
finite impulse response (FIR), 263, 319
FIR vs. IIR, 346-350
frequency domain parameters, 268-270
frequency response, 176-179, 268-270, 562
high-pass, 110, 129, 268-274
low-pass, 110, 128-129, 268-270, 280, 285-288, 343-350, 400-401
matched filter, 138, 307-310, 416-419
moving average filter, 277-284, 307-310, 348-350, 406-407
odd order filter, 603
overshoot, step response, 266-267
passband (defined), 268
passband ripple (defined), 268
pulse response, 328-332
roll-off (defined), 268-269
smoothing, 110, 128-129, 280-282, 348-350
spectral inversion, 270-272, 293
spectral reversal, 273-274
step response, 262-263, 265-267, 338
stopband (defined), 268
stopband attenuation, 268-269, 293-296
time domain parameters, 266-268